



**SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES**



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Analytical Problem

Code of Conduct:

I Ebin Antony P (192373011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

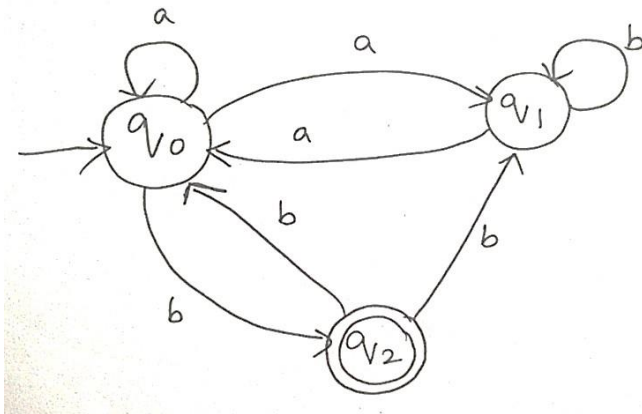
Signature of the student:



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1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma=\{a,b\}$. Construct a DFA that represents the same language L

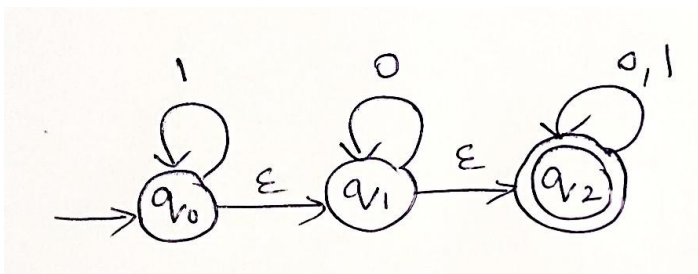


2. Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb , Construct an NFA for L , Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma=\{0,1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states





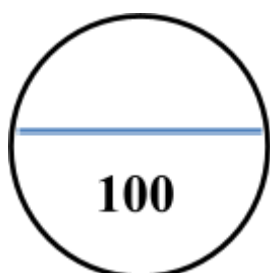
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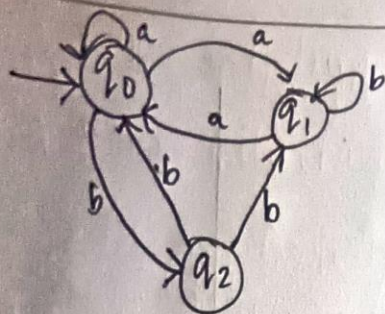


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RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50





- Initial state: q_0
- Final state: q_2
- Alphabet: $\Sigma = \{a, b\}$

NFA Transition Table

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$*q_2$	$\emptyset$	$\{q_0, q_1\}$

($*q_2$ is the accepting state)

Subset Construction (DFA)

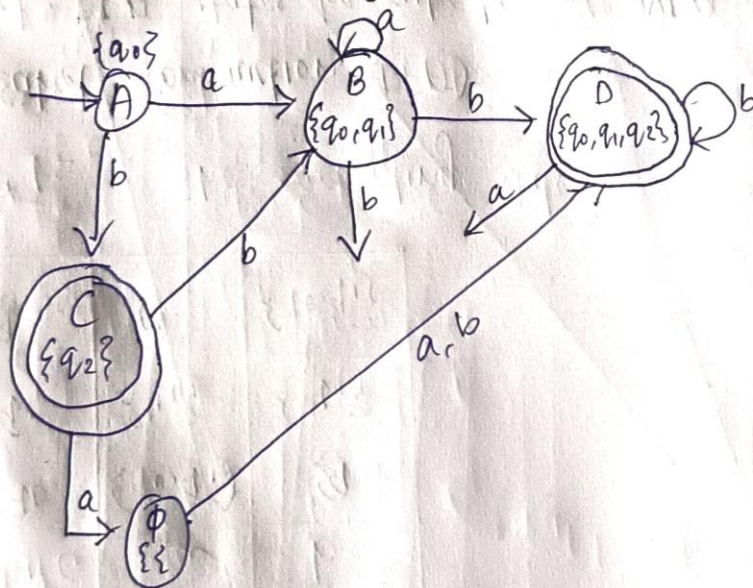
Let DFA states be subsets of NFA states

DFA state	a	b	Final
$A = \{q_0\}$	B	C	No
$B = \{q_0, q_1\}$	B	D	No
$C = \{q_2\}$	$\emptyset$	B	Yes
$D = \{q_0, q_1, q_2\}$	B	D	Yes
$\emptyset \{ \}$	$\emptyset$	$\emptyset$	No

DFA State Naming

- $A = \{q_0\}$ (Start)
- $B = \{q_0, q_1\}$
- $C = \{q_2\}$ (Final)
- $D = \{q_0, q_1, q_2\}$ (Final)
- $\phi = \text{Dead State}$

DFA Transition Diagram



Final DFA

- Start State: $A = \{q_0\}$
- Accepting States: $C = \{q_2\}$, $D = \{q_0, q_1, q_2\}$

This is the DFA equivalent to the given NFA.

Q2. NFA and Equivalent DFA

(2)

Language

Strings over $\Sigma = \{a, b\}$ of length at least 2 whose last two symbols are different.

Accepted: $ab, ba, aab, bba, abaab$

Not accepted: $aa, bb, abbaa$.

NFA

States:

q_0 - Start

q_1 - Last symbol guessed as a .

q_2 - Last symbol guessed as b .

q_3 - Final (ends with ab)

q_4 - Final (ends with ba)

Transition Table

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	—	$\{q_3\}$
q_2	$\{q_4\}$	—
q_3	—	—
q_4	—	—

Final states = $\{q_3, q_4\}$

Equivalent DFA

State	a	b
$\rightarrow S$	A	B
A	AA	AB*
B	BA*	BB
AA	AA	AB*
AB*	BA*	BB
BA*	AA	AB*
BB	BA*	BB

Final States = AB, BA

Q3. NFA for strings ending in 00 or 11

Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

Start: q_0

Final states: $\{q_2, q_4\}$

Transition Function

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_2\}$	$\emptyset$
$*q_2$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\{q_4\}$
$*q_4$	$\emptyset$	$\emptyset$

Acceptance of 01011

q_0

0 $\rightarrow q_0, q_1$

1 $\rightarrow q_0$

0 $\rightarrow q_0, q_1$

1 $\rightarrow q_0$

1 $\rightarrow q_0, q_3, q_4$

Since q_4 is reached, 01011 is accepted

(3)

Q4. DFA for strings containing 001.

Language: All binary strings having 001 as a substring.

DFA States:

- q_0 - Start
- q_1 - Seen 0
- q_2 - Seen 00
- q_3 - Seen 001 (Accept)

Transition Table

State	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$*q_3$	q_3	q_3

Final state = q_3

Q5. E-NFA to DFA.

Given E-NFA

- q_0 has loop on 1
- $q_0 \xrightarrow{\epsilon} q_1$
- q_1 has loop on 0
- $q_1 \xrightarrow{\epsilon} q_2$
- q_2 is final with loop on 0,1.

ϵ -closure

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

DFA States

$A = \{q_0, q_1, q_2\}$ (start)

$B = \{q_1, q_2\}$

$C = \{q_2\}$

All contain final state q_2 , so all are accepting.

DFA Transition Table.

DFA state	0	1
$\rightarrow^* A$	B	A
$* B$	B	C
$* C$	C	A

Start state = A

Final state = A, B, C